

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. – 02

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Recap of Previous Lecture



Topic

Network Address

Topic

Broadcast Address



Topics to be Covered



Topic

Network Size

Topic

Network Mask

Example 4 :-

[MCQ]



#Q. Consider a class-less IPv4 network, the broadcast address (special IPv4 address) of this network is "175 . 175 . 175 . 255" and size of network identifier field is 22 bits, which one of the following can be a network address (special IPv4 address) of this network ?

- A 175 . 175 . 0 . 0
- B 175 . 175 . 160 . 0
- ☒ C 175 . 175 . 172 . 0
- D 175 . 175 . 175 . 0

Ans: C

Network Directed Broadcast Address = 175.175.175.255

10101111 10101111 10101111 11111111
 22 bit NetID 10 bit HostID

Network Address =

10101111 10101111 10101100 00000000
 22 bit NetID 10 bit HostID
 175.175.172.0

Example 5 :-

[MSQ]



#Q. Consider a class-less IPv4 network, the network address (special IPv4 address) of this network is "145 . 75 . 80 . 0", which is/are can be a broadcast address (special IPv4 address) of this network ?

☒ A 145 . 75 . 80 . 0

☐ B 145 . 75 . 80 . 255

☒ C 145 . 75 . 95 . 0

☐ D 145 . 75 . 95 . 255

Ans: B & D

Network Address = 145.75.80.0

10010001 01001011 01010000 00000000

Min^m 20 bit NetID

Max^m 12 bit HostID

Network Directed Broadcast Address =

10010001 01001011 01010000 00000011

30 bit NetID

2 bit HostID

10010001 01001011 01011111 11111111

20 bit NetID

12 bit HostID

$$\times 145.75.80.0 = 145.75.01010000.00000000$$

32 bit NetID

$$\checkmark 145.75.80.255 = 145.75.01010000.11111111$$

24 bit NetID 8 bit HostID

$$\times 145.75.95.0 = 145.75.01011111.00000000$$

20 bit NetID

$$\checkmark 145.75.95.255 = 145.75.01011111.11111111$$

20 bit NetID 12 bit HostID

Example 6 :- [H.W.]

[MSQ]



#Q. Consider a class-less IPv4 network, the broadcast address (special IPv4 address) of this network is "175 . 175 . 175 . 255, which is/are can be a network address (special IPv4 address) of this network ?

- A 175 . 160 . 0 . 0
- B 175 . 175 . 0 . 0
- C 175 . 175 . 160 . 0
- D 175 . 175 . 175 . 0



Topic : Multicast Transmission



* IGMP

→ One-to-many transmission

→ One host is sender and group of hosts are receiver

Source IP Address = Host IP Address

Destination IP Address = Multicast Address

[Special IP Address as a Group ID]



Topic : This Host

- Special IP address (all zero bits)
- Unspecified IP Address
- This Host : 0.0.0.0



Topic : Host IP Address

- Host IP address
- Used to identify a host uniquely world wide

NetID field (x bits) = [As Assigned]

HostID field (y bits) = [Any thing]

[Except all zero and all one bits]

Example 7 :-

#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "1100 1100 1010", what is the first and last IP address of this network which can be assigned to a host?

Network Address =

11001100 10100000 00000000 00000000
 12 bit NetID 20 bit HostID

First IP Address that can be assigned to a host =

11001100 10100000 00000000 00000001
 20 bit HostID

Last IP Address that can be assigned to a host =

11001100 10101111 11111111 11111110

$(2^{20} - 2)$ hosts

Network Directed Broadcast Address =

11001100 10101111 11111111 11111111
 NetID HostID

Network Address

$$= 204 \cdot 160 \cdot 0 \cdot 0$$

First IP Address

[that can be assigned to a host]

$$= 204 \cdot 160 \cdot 0 \cdot 1$$

⋮

Last IP Address

[that can be assigned to a host]

$$= 204 \cdot 175 \cdot 255 \cdot 254$$

Network Directed Broadcast Address

$$= 204 \cdot 175 \cdot 255 \cdot 255$$

Example 8 :-

#Q. Consider an IPv4 address of a class-less IPv4 network is "115 . 120 . 125 . 130 / 19", write network address, broadcast address, first and last IP address of this network which can be assigned to a host ?

IPv4 Address = $115.120.125.130/19$

= $115.120.01111101\ 100000010$
 19 bit NetID 13 bit HostID

Network Address = $115.120.96.0/19$

First IP Address
 [that can be assigned to a host]

= $115.120.96.1/19$

Last IP Address
 [that can be assigned to a host]

= $115.120.127.254/19$

$(2^{13} - 2)$
 hosts

Network Directed Broadcast Address = $115.120.127.255/19$

Network Address =

115.120.01100000 00000000
 19 bit NetID 13 bit HostID

First IP Address that can be assigned to a host =

115.120.01100000 00000001
 NetID HostID

Last IP Address that can be assigned to a host =

115.120.01111111 11111110
 HostID

Network Directed Broadcast Address =

115.120.01111111 11111111
 HostID

[NAT]

[GATE-2015][2 Mark]



#Q. In the network "200 . 10 . 11 . 144 / 27", the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.

144 $\rightarrow$ 100 | 0000
NetID 5 bit HostID

Ans = 100 | 11110 = 158
NetID 5 bit HostID

Ans = 158

IPv4 Address = 200.10.11.144 / 27

= 200.10.11.10010000
 27 bit NetID 5 bit HostID

Last IP Address that can be assigned to a host =

200.10.11.10011110
 27 bit NetID 5 bit HostID

Last IP Address = 200.10.11.158
 [that can be assigned to a host]



Topic : Network Size

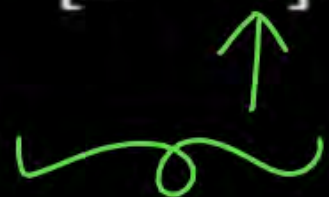


→ Network Size : Maximum number of hosts can be in a network

→ HostID field (y bits)

$$[y \geq 2]$$

Network Size = $[2^y - 2]$ hosts per network



Example 9 :-

[NAT]



#Q. Consider a class-less IPv4 network, the number of bits in the network id field are 22, how many maximum number of hosts can be in this network?

$$\begin{aligned}\text{Ans} &= [2^{10} - 2] \text{ hosts} \\ &= 1022\end{aligned}$$

Number of bits in Network ID field (x) = 22 bits

Number of bits in Host ID field (y) = (32 - x) = (32 - 22) = 10 bits

Maximum number of hosts can be in a network = $[2^y - 2]$ hosts per network

$$= [2^{10} - 2] \text{ hosts}$$

$$= 1022$$



Topic : Network Mask



→ Network Mask (Netmask) [32-bits]

NetID field = All One Bits

HostID field = All Zero Bits

Network Id [1 1 1 1 1 1 1]

Host Id [0 0 0 0 0 0 0]

x bit

y bit



Topic : Network Mask



→ Binary string used to produce “Network Address” of an IP address

$$\text{Network Address} \leftarrow \text{IP Address} [\text{bit-wise AND}] \text{ Netmask}$$

→ All hosts belongs to same network must have same “Network Address”



Topic : Network Mask



IP Address :
[bit wise AND]

Network Mask :

Network Id	Host Id
------------	---------

[bit - wise AND]

1 1 1 1 1 1 1	0 0 0 0 0 0 0
---------------------	---------------------

Network Address :

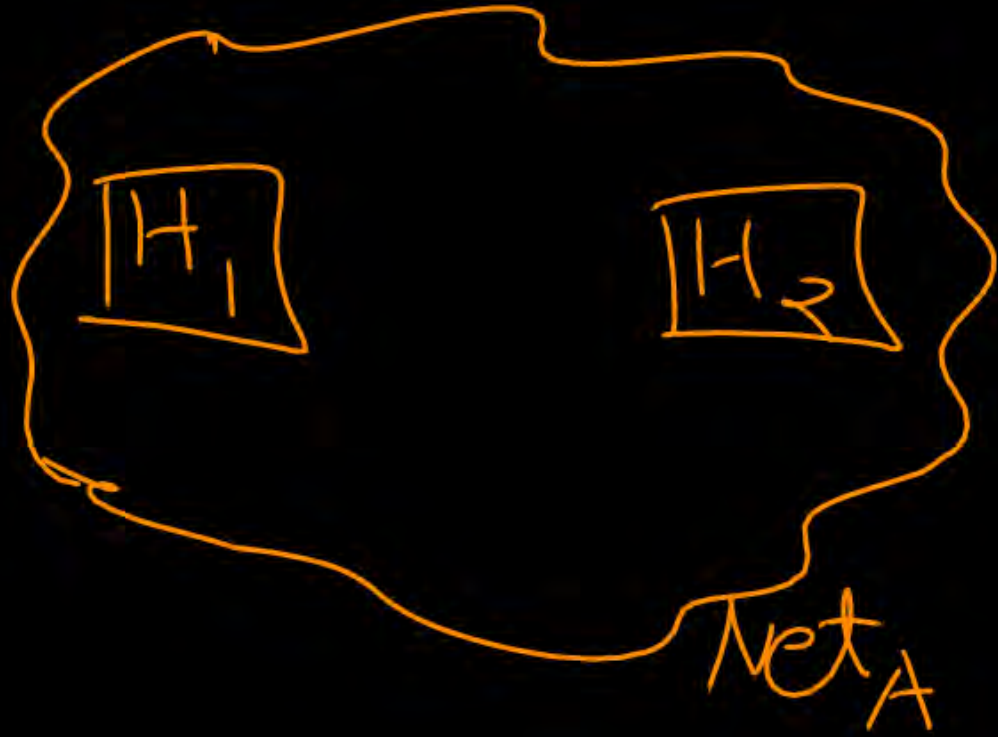
Network Id	0 0 0 0 0 0 0
------------	---------------------

x bit

y

$$a \text{ AND } 1 = a$$

$$a \text{ AND } 0 = 0$$



$$\begin{array}{r} IP_{H1} \\ \text{NetMask} \\ \hline \text{NetAdd}_A \end{array}$$

$$\begin{array}{r} IP_{H2} \\ \text{NetMask} \\ \hline \text{NetAdd}_A \end{array}$$



$$\begin{array}{r} IP_{H3} \\ \text{NetMask} \\ \hline \text{NetAdd}_B \end{array}$$

Example 10 :-

[MCQ]



#Q. Consider a class-less IPv4 network where the size of network identifier field is 22 bits, which one of the following is network mask of this network ?

- A 255.255.0.0
- B 255.255.248.0
- ✓ C 255.255.252.0
- D 255.255.255.0

Ans: C

Network Mask = 11111111 11111111 11111100 00000000

22 bit NetID 10 bit HostID

= 255.255.252.0

Example 11 :-

[NAT]



#Q. Consider a class-less IPv4 network, the network mask of this network is "255 . 255 . 248 . 0", how many maximum number of hosts can be in this network ?

$$\begin{aligned} \text{Ans} &= (2^{11} - 2) \text{ hosts} \\ &= 2046 \text{ hosts} \end{aligned}$$

Network Mask = 255 . 255 . 248 . 0

= 11111111 11111111 11111000 00000000
Net ID
11 bit Host ID

Number of bits in Host ID field (y) = 11

Maximum number of hosts can be in a network = $[2^y - 2]$ hosts per network
 $= (2^{11} - 2) \text{ hosts}$

Example 12 :-

[MSQ]



#Q. Consider two host belongs to a class-less IPv4 network, the IP addresses of the hosts are "178 . 158 . 138 . 118" and "178 . 168 . 158 . 148", which is/are can be a network mask of this network ?

☒ A 255 . 0 . 0 . 0

☒ B 255 . 192 . 0 . 0

☐ C 255 . 240 . 0 . 0

☐ D 255 . 255 . 0 . 0

Ans: A & B



178.158.138.118

255. 0. 0. 0

178. 0. 0. 0

← Same →

178.168.158.148

255. 0. 0. 0

178. 0. 0. 0

178 → 10110010

255 → 11111111

178 ← 10110010

178.158.138.118

255.192. 0. 0

178.128. 0. 0

← Same →

178.168.158.148

255.192. 0. 0

178.128. 0. 0

118 → 01110110

0 → 00000000

0 ← 00000000

158 → 10011110

192 → 11000000

128 ← 10000000

168 → 10101000

192 → 11000000

128 ← 10000000

178.158.138.118

255.240. 0. 0

178.144. 0. 0

178.168.158.148

255.240. 0. 0

178.168. 0. 0

Not Same

158 → 10011110

240 → 11110000

144 ← 10010000

178.158.138.118

255.255. 0. 0

178.158. 0. 0

178.168.158.148

255.255. 0. 0

178.168. 0. 0

Not Same

168 → 10101000

240 → 11110000

160 ← 10100000

$$178.158.138.118 = \underline{178}.10011110.138.118$$

$$178.168.158.148 = \underline{178}.10101000.158.148$$

$$255.0.0.0 = \underline{255}.00000000.0.0$$

$$255.192.0.0 = 255.\underline{11}000000.0.0$$

$$255.240.0.0 = 255.11110000.0.0$$

$$255.255.0.0 = 255.11111111.0.0$$

[MCQ]

11T-6, H.W.

[GATE-2010][2 Mark]



#Q. Suppose computers A and B have IP addresses "10 . 105 . 1 . 113" and "10 . 105 . 1 . 91" respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

- A 255 . 255 . 255 . 0
- B 255 . 255 . 255 . 128
- C 255 . 255 . 255 . 192
- D 255 . 255 . 255 . 224

[MCQ]

IIT-M, H.W.

[GATE-2003][2 Mark]



netmask

#Q. The subnet mask for a particular network is "255 . 255 . 31 . 0". Which of the following pairs of IP addresses could belong to this network ?

- (A) 172 . 57 . 88 . 62 and 172 . 56 . 87 . 233
- (B) 10 . 35 . 28 . 2 and 10 . 35 . 29 . 4
- (C) 191 . 203 . 31 . 87 and 191 . 234 . 31 . 88
- (D) 128 . 8 . 129 . 43 and 128 . 8 . 161 . 55



Topic : IPv4 Address



Problem : How many bits are assigned for network id field ?

→ IPv4 address are categorized into two type :

1. Classful IPv4 Address

→ Static Assignment : NetID field size defined implicitly

2. Classless IPv4 Address

→ Dynamic Assignment : Need to define NetID field size explicitly

↓
Slash or prefix notation



Topic : Classful IPv4 Address

→ Classful IPv4 Address are of five types :

1. Class A

2. Class B

3. Class C

4. Class D

5. Class E



2 mins Summary



Topic

Network Size

Topic

Network Mask



THANK - YOU

